

Infrared SpectroscopyPrinciple[Vibrational spectroscopy]

IR spectrum is an important record which gives sufficient information about structure of compound. Infrared spectrum of a molecule results due to transitions two different vibrational energy levels. The vibrational motion resembles the motion of a simple harmonic oscillator. The vibrational energy of a chemical bond is quantised and can have the value $E_{\text{vib}} = (v + \frac{1}{2}) h\nu$

v is vibrational energy level 0, 1, 2, 3, ...

h is Planck's constant

When $v = 0$ $E_{\text{vib}} = (0 + \frac{1}{2}) h\nu = \frac{1}{2} h\nu$.

where This is called zero point energy which signifies that a molecule must always vibrate and can never be at rest.

The vibrational motion of a diatomic molecule is similar to vibration of a simple harmonic oscillator.

Frequency of S.H.M is given by $\nu = \frac{1}{2\pi} \sqrt{\frac{k}{M}}$

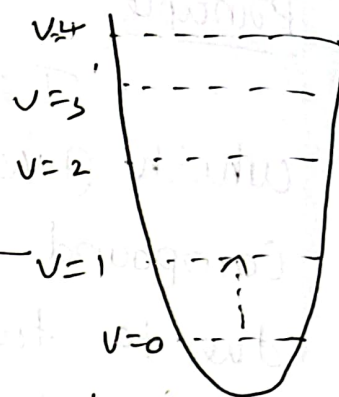
Where k = force constant

$$\gamma = \text{reduced mass} = \frac{m_1 m_2}{m_1 + m_2}$$

In spectroscopy, the unit commonly used is wave number $\bar{\nu} = \frac{\nu}{\epsilon}$

(Vibrational Frequency)

$$\bar{\nu} = \frac{1}{2\pi c} \sqrt{\frac{k}{\gamma}}$$



Requisite for IR Spectra

All type of molecules can not interact with IR radiation. Only those molecules which can exhibit change in dipole moment during vibration can exhibit IR spectrum. Homonuclear diatomic molecules (H_2 , O_2 , N_2 or Cl_2) do not exhibit IR spectrum. HCl , HBr , and HI molecules are IR active. Since these molecules exhibit change in dipole moment during vibration.

Number of Vibrational modes in a molecule.

For a linear molecule

$$\text{Vibrational modes} = 3n - 5$$

where 'n' is the number of atoms in a molecule.

For a nonlinear molecule (3)

$$\text{Vibrational modes} = 3n - 6$$

Draw the vibrational modes CO_2 , H_2O and HCl molecule.

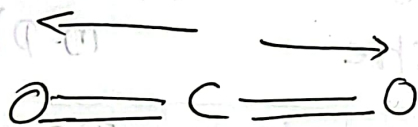
CO_2

For linear molecule

$$\text{no. of vibrational modes} = 3n - 5$$

$$= 3 \times 3 - 5$$

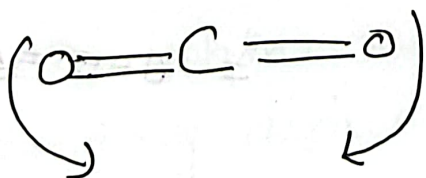
$$= 9 - 5 = \underline{\underline{4}}$$



Symmetric stretching

No change in dipole moment

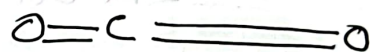
IR inactive



Bending in plane

Change in dipole moment

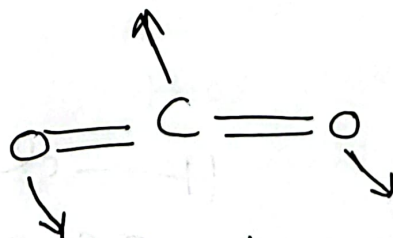
IR active



Asymmetric stretching

Change in dipole moment

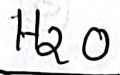
IR Active



Bending out plane

Change in dipole moment

IR active



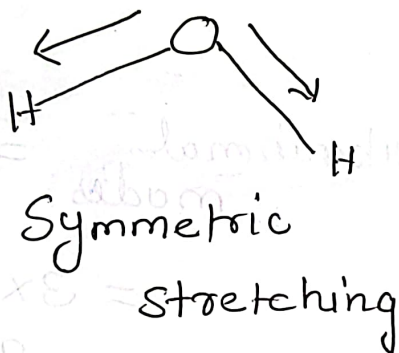
Nonlinear molecule

$$\text{Vibrational modes} = 3n - 6$$

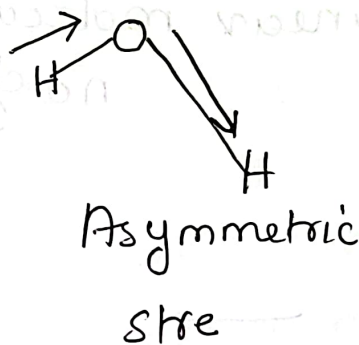
$$n = 3$$

$$= 3 \times 3 - 6$$

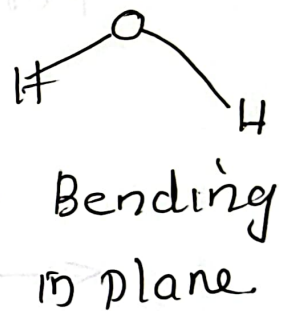
$$= 9 - 6 = \underline{\underline{3}}$$



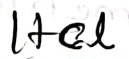
IR active



IR active



IR active



$$\text{Vibrational modes} = 3n - 5$$

$$= 3 \times 2 - 5$$

$$= 6 - 5 = \underline{\underline{1}}$$



IR active

Change in dipole moment.

Applications of IR Spectroscopy

(1) To identify the functional group of a molecule.

Functional group	IR Frequency
-OH	3200-3500 cm^{-1}
-C $\equiv$ C-	2000-2200 cm^{-1}
-C=C-	1600-1680 cm^{-1}

(2)

(2) Determination of Purity. IR spectra of impure sample show some extra-absorption bands.

3. To distinguish between intramolecular and intermolecular hydrogen bonding.

This can be done in taking a series of IR spectra of the compound at different dilutions. As the dilution is increased, the absorption band due to intermolecular H-bonding diminishes while that due to intramolecular H-bonding remains unchanged.

(4) Determination of force constant

$$\bar{\nu} = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}} \quad \bar{\nu}^2 = \frac{1}{4\pi^2 c^2} \frac{k}{\mu}$$

$$\therefore k = \bar{\nu}^2 \times 4\pi^2 \times c^2 \times \mu$$